

ApiaryLens Product Overview and Capability Whitepaper

Family, friends, bee clubs, and contributor overview

Generated: 2026-07-07 03:27:20

ApiaryLens Product Overview and Capability Whitepaper

What is ApiaryLens?

ApiaryLens is an open-source apiary intelligence platform for beekeepers. It helps track hives, queens, inspections, hive health, photos, weather, blooms, honey production, and long-term colony history.

ApiaryLens is designed to be simple enough for a family getting its first hive and powerful enough to grow into bee club, research, and commercial use.

Why It Exists

New beekeepers need more than a notebook. They need a way to remember what happened in each hive, understand seasonal tasks, track queen performance, watch for health issues, and learn from mentors.

ApiaryLens turns hive records into a living history.

Who It Is For

- New beekeepers
- Families
- Bee clubs
- Mentors
- Commercial apiaries
- Extension offices
- Educators
- Researchers

MVP Capabilities

The first major version should focus on practical hive management.

MVP features:

- Multi-apiary tracking
- Multi-hive tracking
- Hive equipment tracking
- Box and frame records
- Queen tracking
- Inspection logging

- Photos and videos
- Hive health observations
- Varroa mite tracking
- Feeding records
- Treatment records
- Honey harvest records
- Weather history
- Bloom calendar foundation
- QR codes
- Sharing with family or mentors
- PWA support
- Offline-first design
- Docker self-hosted deployment
- REST API

Future Capabilities

ApiaryLens should grow into a full apiary intelligence platform.

Future features:

- AI photo review
- AI inspection summaries
- Brood pattern analysis
- Queen detection
- Swarm risk flags
- Honey flow forecasting
- Native iOS and Android apps
- Bee club mode
- Research mode
- Plugin marketplace
- Home Assistant integration
- Hive scales and sensors
- Route planning
- Commercial reporting
- Optional hosted SaaS

Photos and Video Almanac

ApiaryLens should preserve the visual story of each hive.

A beekeeper should be able to look back over time and see:

- What the brood looked like
- How the queen performed
- When pests appeared
- How honey stores changed
- What the entrance looked like
- How the hive responded through seasons

This creates a practical photo and video almanac for each hive.

Weather and Bloom Intelligence

ApiaryLens should connect hive performance to the local environment.

The system should track:

- Historical weather
- Rainfall
- Heat and cold events
- Frost dates
- Bloom timing
- Nectar and pollen sources
- Weather impact on inspections
- Weather impact on honey production

Open Source and Self-Hosted

ApiaryLens should be free or as close to free as possible for families and bee clubs.

Core principles:

- No required paid cloud account
- No required AI subscription
- Runs locally or on a small server
- Docker Compose deployment
- User owns the data
- Exportable data

Future SaaS Possibility

ApiaryLens is not SaaS at launch.

But the architecture should allow a future hosted version for users who want easy onboarding and do not want to self-host.

The open-source project remains first-class even if a hosted version exists later.

Brand Positioning

ApiaryLens means seeing the hive clearly.

Possible tagline:

> ApiaryLens: Open-source insight for every hive.

Alternative tagline:

> ApiaryLens: See every hive. Understand every colony.